

EXPERIMENT 22 — Candidate-Elimination Algorithm

Aim

To implement the Candidate-Elimination algorithm and determine the S (specific) and G (general) boundaries of the version space for a given training dataset.

Algorithm / Procedure

1. Read the training dataset from enjoysport.csv.
2. Initialize S with the first positive example.
3. Initialize G with the most general hypothesis (? for every attribute).
4. For every training example:
 - If the example is positive, generalize S where required and remove inconsistent hypotheses from G.
 - If the example is negative, specialize G so that it excludes the negative example.
5. Remove hypotheses that are inconsistent with the training examples.
6. Display the final S and G boundaries.

CODE

```
import pandas as pd

def more_general(h1, h2):
    return all(x == '?' or x == y for x, y in zip(h1, h2))

def candidate_elimination(X, y):
    n = X.shape[1]
```

```

# Initial boundaries
S = [['0'] * n]
G = [['?'] * n]

for i, x in enumerate(X):

    if y[i] == 'Yes':

        # Generalize S
        for j in range(n):
            if S[0][j] == '0':
                S[0][j] = x[j]
            elif S[0][j] != x[j]:
                S[0][j] = '?'

        # Remove G hypotheses that are not general enough
        G = [g for g in G if more_general(g, x)]

    else:

        # Specialize G for negative example
        new_G = []

        for g in G:
            if more_general(g, x):
                for j in range(n):
                    if g[j] == '?':
                        if S[0][j] != x[j]:
                            new_g = g.copy()
                            new_g[j] = S[0][j]
                            new_G.append(new_g)
            else:

```

```

        new_G.append(g)

    G = new_G

    return S, G

# Read dataset
data = pd.read_csv("enjoysport.csv")

X = data.iloc[:, :-1].values
y = data.iloc[:, -1].values

# Apply Candidate-Elimination
S, G = candidate_elimination(X, y)

print("Final S:", S)
print("Final G:", G)

```

OUTPUT

```

... Final S: [['Sunny', 'Warm', '?', 'Strong', '?', '?']]
    Final G: [['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?']]

```

Result

The Candidate-Elimination algorithm was successfully implemented, and the S and G boundaries of the version space were obtained from the given training dataset.